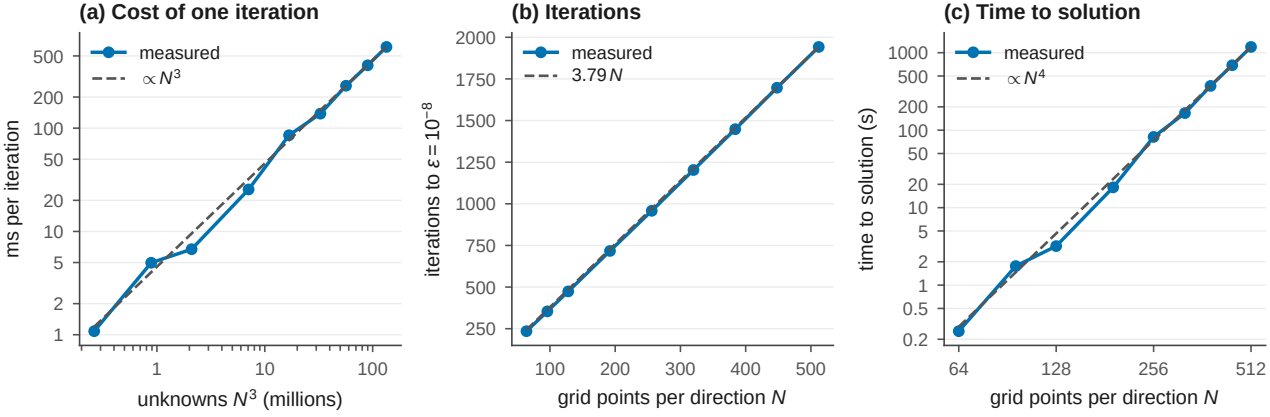
(a) Cost of one iteration
measured
$\propto N^3$
ms per iteration
500
200
100
50
20
10
5
2
1
1
10
100
unknowns N^3 (millions)
(b) Iterations
measured
3.79 N
iterations to $\varepsilon = 10^{-8}$
2000
1750
1500
1250
1000
750
500
250
100
200
300
400
500
grid points per direction N
(c) Time to solution
measured
$\propto N^4$
time to solution (s)
1000
500
200
100
50
20
10
5
2
1
0.5
0.2
64
128
256
512
grid points per direction N